

Introduction to Biostatistics (MSBD520) - Fall 2026
Instructor: Nazirah Mohd Khairi, PhD

Sampling, Experimental Design, and Inference

The Methodological Filters of Data Collection

Course Dashboard



Office Hours

Wednesdays:
11:00 am – 12:00 pm CT

Fridays:
3:00 pm – 4:00 pm CT



Homework

Assignment: HW1 is posted
on Blackboard.

Due Date: Next week
(September 3).



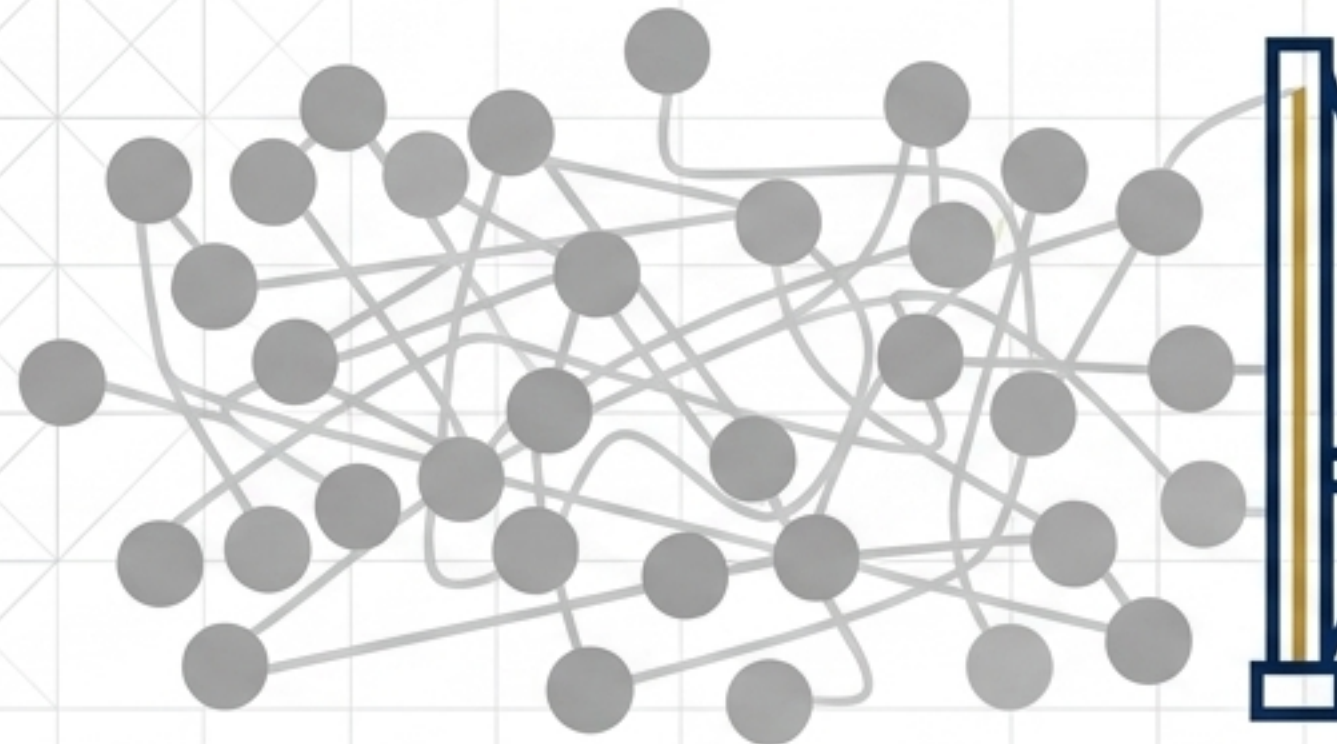
Deadlines & Penalties

Today: In-class work must
be uploaded by midnight.

Alert: Starting next week,
any late assignments will
incur a grade penalty.

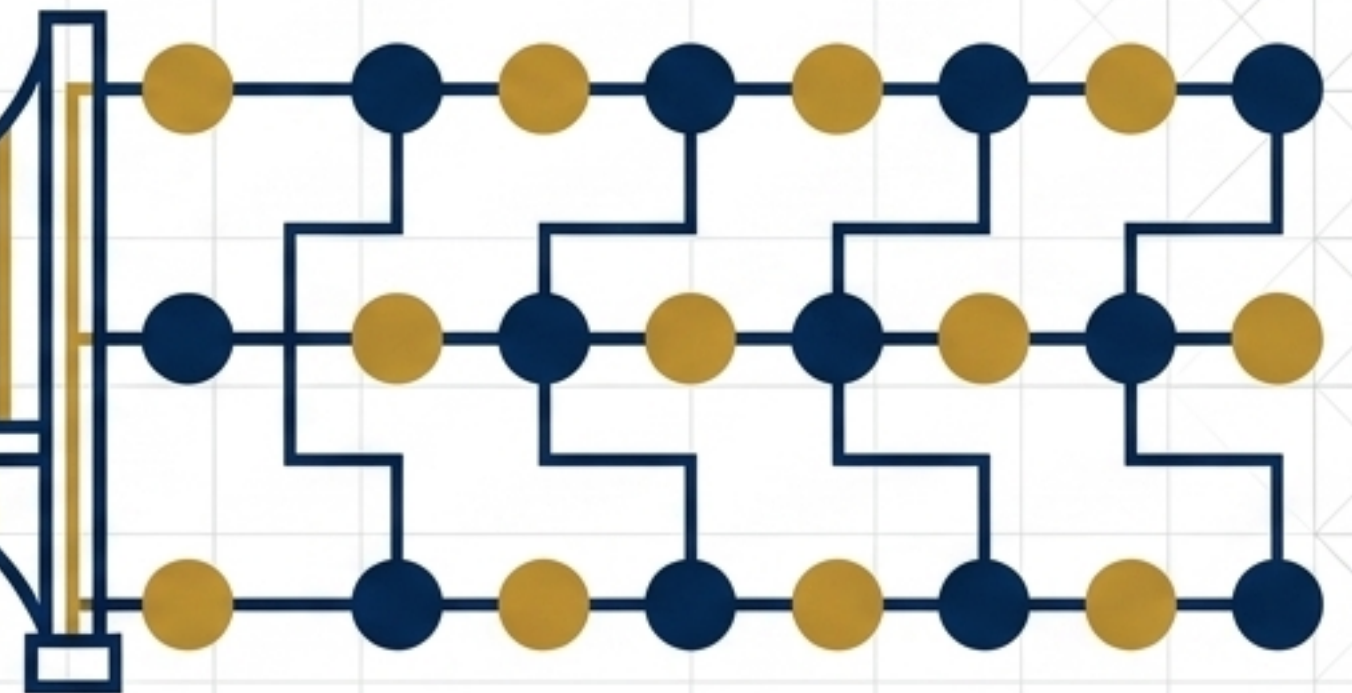
Roadmap: Where We Are

The Past



Data matrices, Variable types, Relationships,
Basic data collection & sampling

Today

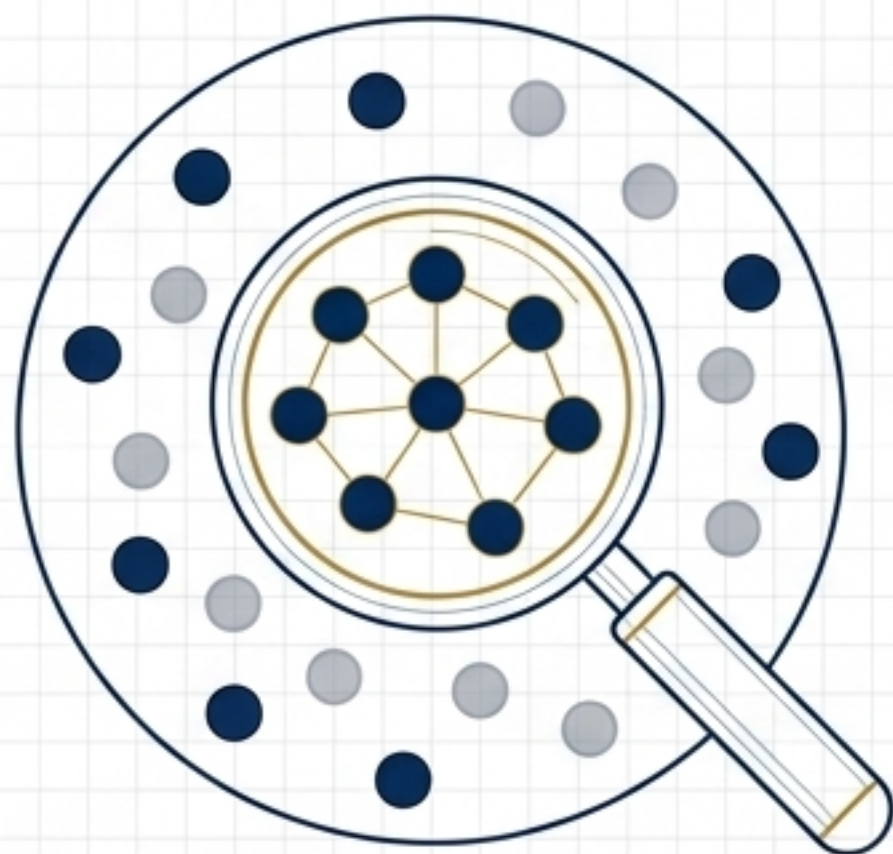


Principles of experimental design, Mitigating
bias, Establishing causal inference

Moving from collecting data to controlling environments.

The Threat: Three Pitfalls of Sampling

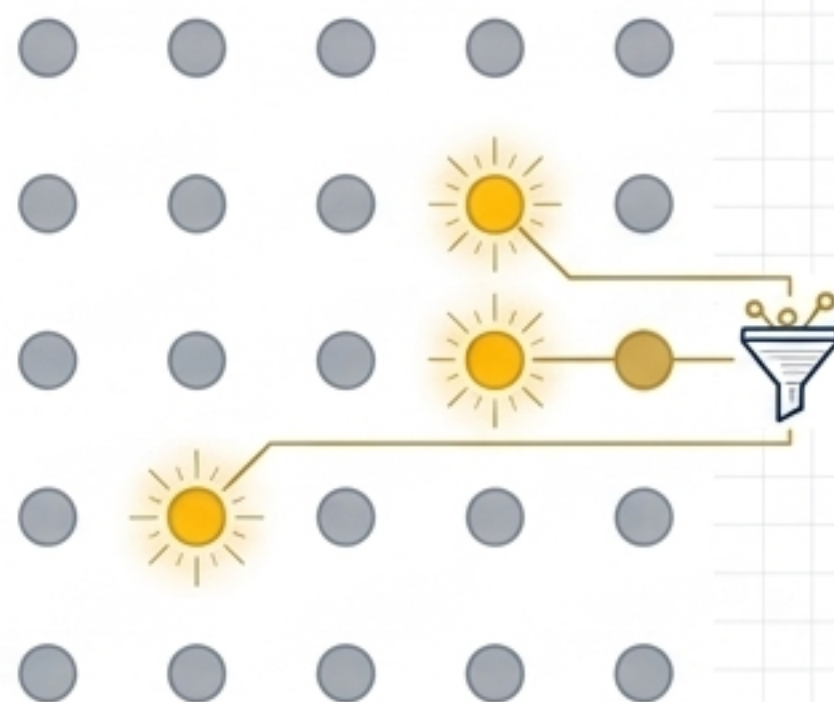
Sampling Bias



The Pick and Choose Error

Selective inclusion creates a fundamentally skewed population.

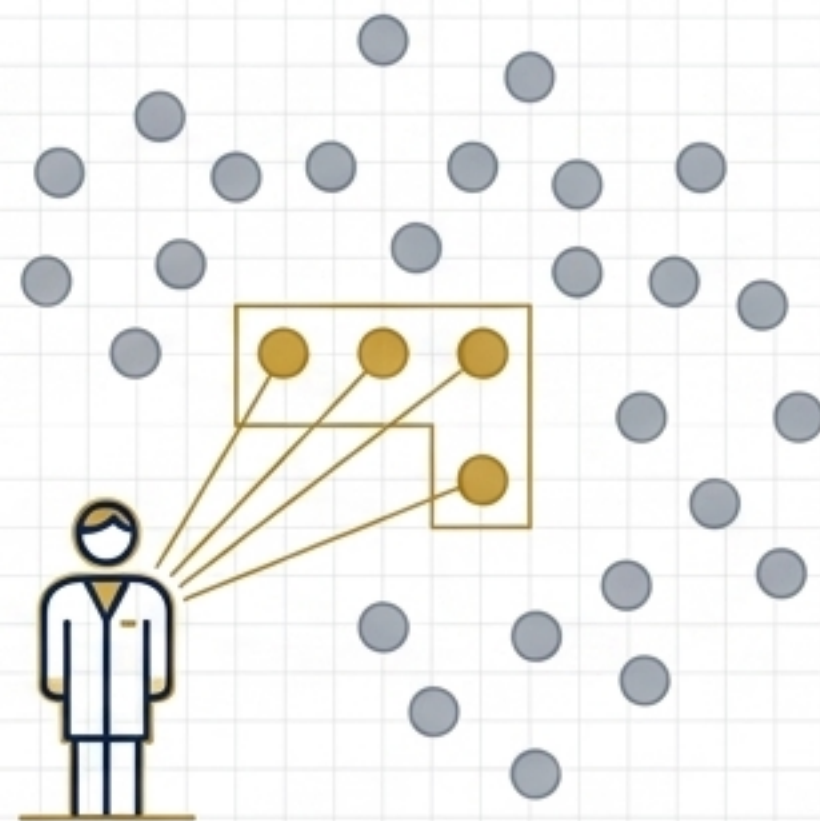
Non-Response Bias



The Silent Majority Error

If only a small fraction of a random sample responds, the resulting data is no longer representative.

Convenience Sampling



The Path of Least Resistance Error

Over-indexing on individuals who are easily accessible.

The Four Pillars of Experimental Design

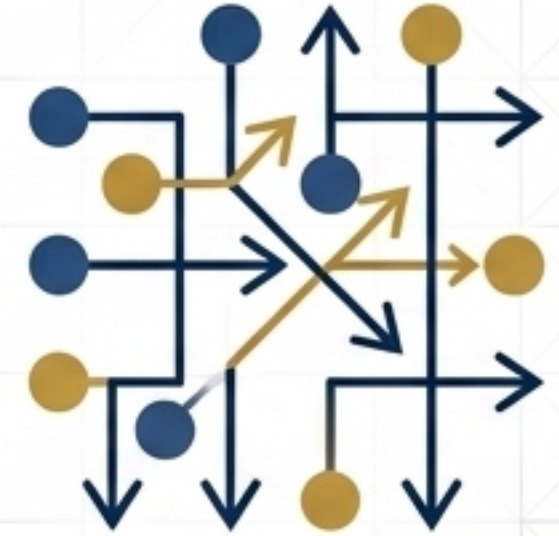
1. Control

Control for the potential effect of variables other than the ones directly being studied.



2. Randomize

Randomly assign subjects to treatments, and randomly sample from the population whenever possible.



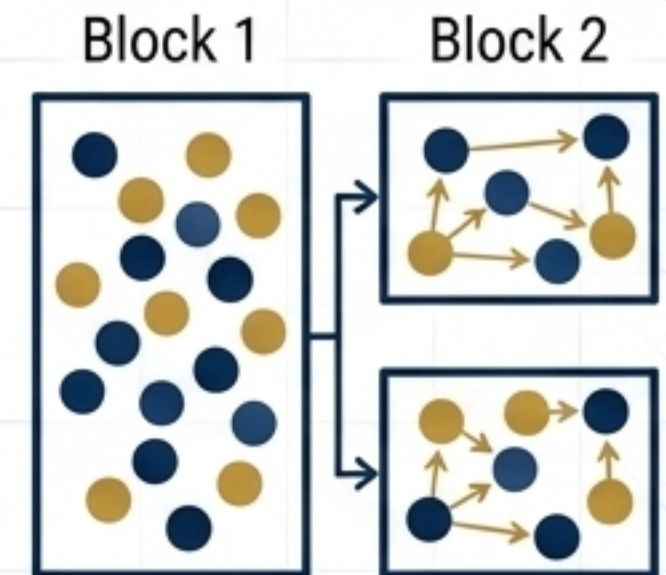
3. Replicate

Within a study, replicate by collecting a sufficiently large sample. Or replicate the entire study entirely.



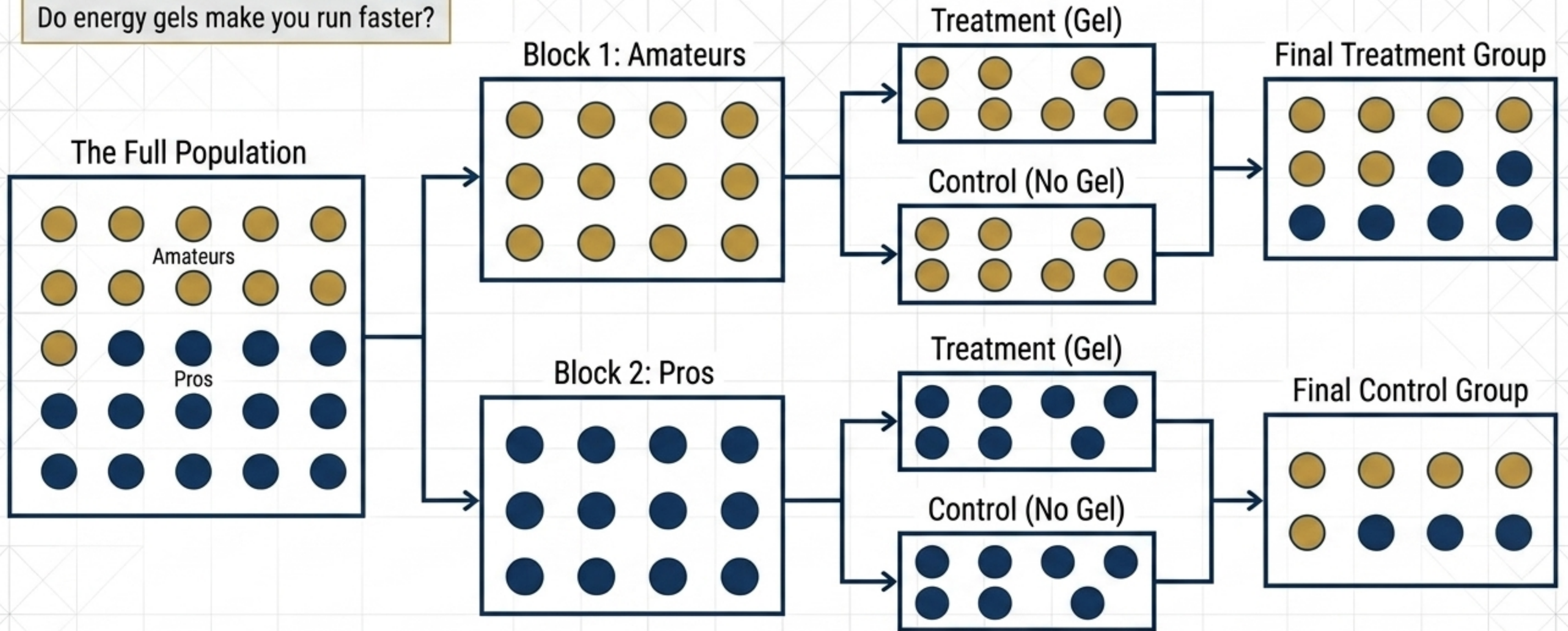
4. Block

If there are suspected influential variables, first group subjects into blocks based on these variables, then randomize cases within each block.



Deep Dive: The Mechanics of Blocking

Do energy gels make you run faster?



Visual Proof: Pro/Amateur status is perfectly equalized across both the treatment and control groups.

Comparison Matrix: Factors vs. Blocks

Explanatory Variables (Factors)	Blocking Variables
Conditions we actively IMPOSE on experimental units (e.g., giving the energy gel). This is the core focus of the test.	Characteristics the experimental units COME WITH (e.g., pro status). Used purely to control variance.

(Note: Blocking is to experiments what stratifying is to sampling).

Blueprint in Practice

A study tests the effect of light level and noise level on exam performance. The researcher suspects gender changes the effect, so both genders are equally represented in each group.

Explanatory (Imposed):
Light Level, Noise Level (2)

Blocking (Pre-existing):
Gender (1)

Response (Outcome):
Exam Performance (1)

Controlling the Human Element: Placebos & Blinding

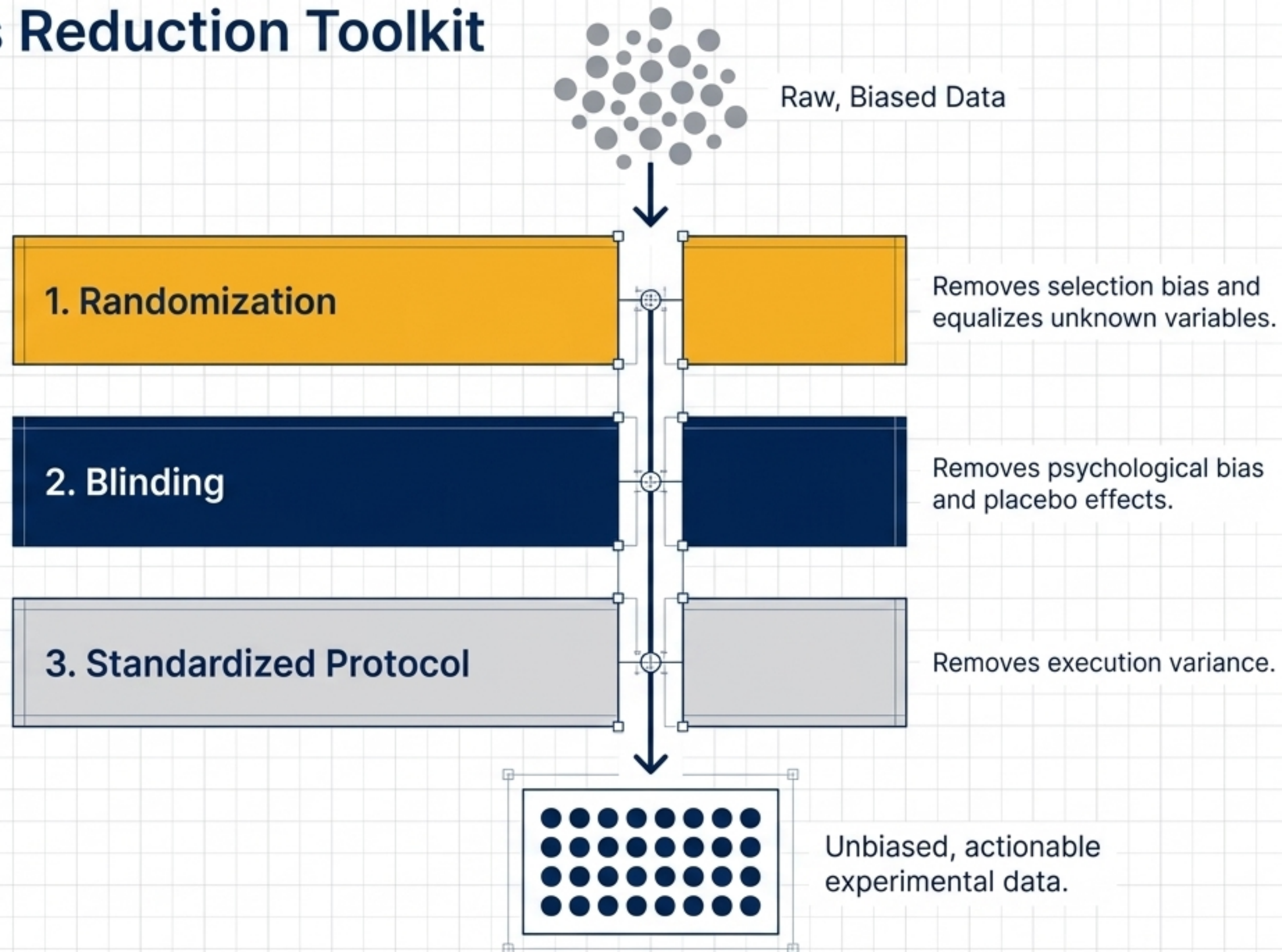
Information Visibility Matrix	Patient Knows Treatment?	Researcher Knows Treatment?
Single-Blind	 No	 Yes
Double-Blind	 No	 No

Double-blind prevents researchers from inadvertently telegraphing information or grading responses differently.

Placebo: A fake treatment used as a baseline control.

Placebo Effect: Improvement shown simply because units believe they are receiving special treatment.

The Bias Reduction Toolkit



SYNTHESIS: The Limits of Inference

	Random assignment Dictates Causation	No random assignment Dictates Correlation
Random sampling Dictates Generalizability	Causal conclusion, generalized to the whole population. Ideal Experiment	No causal conclusion, correlation statement generalized to the whole population. Most Observational Studies
No random sampling Dictates No generalizability	Causal conclusion, only for the sample. Most Experiments	No causal conclusion, correlation statement only for the sample. Bad Observational Studies

Case Study Deconstruction: Chia Seeds

In one 2009 study, a team of researchers recruited 38 men and divided them randomly into two groups: treatment or control. They also recruited 38 women, and they randomly placed half of these participants into the treatment group and the other half into the control group. One group was given 25 grams of chia seeds twice a day, and the other was given a placebo. The subjects volunteered to be a part of the study.

Type: Experiment (Researchers actively imposed a treatment)

Conditions: Treatment (25g seeds twice/day) vs. Control (Placebo)

Block Used: Gender (Men and women were randomized separately)

Blinding: Implied (Use of a placebo implies participants don't know their group)

Synthesis Verdict:
Causal? YES (due to random assignment).
Generalizable? NO (they were volunteers, not a random sample).



Group Case Studies

1. Download: Access the group case document via Blackboard.

2. Discuss: Work with your peers to deconstruct the methodology (15 minutes).

3. Synthesize: Reconvene as a class to present your group's findings.

Apply the filters. Identify the biases.